

In addition to the metrics used by the original paper, this project also integrates a suite of additional evaluation metrics to assess model quality and optimise model performance in these areas. All metrics are calculated on both training and test sets to also evaluate potential overfitting, help diagnose model behaviour and guide architectural improvements.

Accuracy and Forecasting Performance Metrics

<i>Metric</i>	<i>Calculation Method</i>	<i>Relevance</i>
Quantile Loss (Pinball Loss)	$\text{mean}(\max(q \times \text{error}, (q - 1) \times \text{error}))$ for quantiles $q = 0.1, 0.5, 0.9$	Evaluates uncertainty quantification at different probability levels, allowing security teams to understand prediction confidence intervals. The 10th and 90th percentiles provide worst-case and best-case scenarios for resource allocation, whilst the median (50th percentile) offers robust central tendency estimation for operational planning.

Binary Classification and Detection Quality Metrics

<i>Metric</i>	<i>Calculation Method</i>	<i>Relevance</i>
AUC-ROC (Area Under Receiver Operating Characteristic Curve)	Area under curve plotting True Positive Rate vs False Positive Rate across classification thresholds	Measures the model's ability to distinguish between threat and non-threat events across all possible decision thresholds. Values approaching 1.0 indicate excellent discriminative capability, essential for binary threat detection scenarios. Independent of class imbalance, making it suitable for cybersecurity datasets where attacks are typically rare events.
False Positive Rate (FPR)	$\text{FP} / (\text{FP} + \text{TN})$	Quantifies operational efficiency by measuring the proportion of benign events incorrectly flagged as threats. High FPR directly increases analyst workload, leads to alert fatigue, and potentially causes security teams to ignore legitimate threats, a critical consideration for maintaining SOC (Security Operations Centre) effectiveness and regulatory compliance under frameworks like DORA.
Alert Precision	$\text{TP} / (\text{TP} + \text{FP})$	Indicates operational reliability by measuring the percentage of alerts that correspond to genuine threats. High precision reduces wasted investigative effort and maintains analyst trust in the system. Critical for stakeholder confidence and justifying resource investment in proactive threat forecasting capabilities.
Attack Coverage	$(\text{Number of correctly predicted incidents}) / (\text{Total incidents}) \times 100$	Measures comprehensive threat awareness by determining what percentage of real-world attacks the model successfully predicts.

Low coverage indicates blind spots in the forecasting model, potentially leaving organisations vulnerable to specific threat categories or attack vectors.

Essential for evaluating the model's effectiveness in shifting from reactive to proactive cybersecurity posture.

Temporal Metrics

Metric	Calculation Method	Relevance
Directional Accuracy (DA)	(Number of correctly predicted trend directions) / (Total predictions) × 100	Measures the percentage of time the model correctly predicts whether threat activity will increase or decrease. Critical for identifying threat escalation patterns and enabling preemptive resource allocation. Particularly relevant for detecting APT campaigns that show gradual intensification over time.
Temporal Correlation Coefficient	Pearson correlation between predicted and actual time-series values: $\text{corr}(\text{predictions}, \text{actuals})$	Assesses how well the model captures temporal dependencies and patterns in threat evolution. High temporal correlation indicates the model understands cyclical patterns, seasonal variations, and trending behaviours in threat landscapes, essential for long-term forecasting accuracy.

Statistical Distribution and Data Characteristics Metrics

Metric	Calculation Method	Relevance
Kolmogorov-Smirnov (K-S) Test	$\max(\text{CDF}_{\text{predicted}}(x) - \text{CDF}_{\text{actual}}(x))$	Useful for detecting data drift and measuring how well the model can discriminate between two classes.
Jensen-Shannon Divergence	$0.5 \times (\text{KL}(P \parallel M) + \text{KL}(Q \parallel M))$, where $M = 0.5 \times (P + Q)$ is the mixture distribution of P and Q	Useful for detecting data drift.

Model Robustness and Generalisation Metrics

Metric	Calculation Method	Relevance
Adversarial Robustness Score	$1.0 / (1.0 + \text{mean}(\text{predictions}_{\text{noisy}} - \text{predictions}_{\text{original}}))$	Shows the model's resilience to adversarial examples, so if malicious actors try to manipulate the model with adversarial inputs, a high score indicates that they are likely to be unsuccessful.

Operational and Deployment Metrics

Metric	Calculation Method	Relevance
Resource Efficiency	$(\text{CPU-seconds} + \text{RAM_GB} \times \text{scaling_factor}) / 1000_{\text{predictions}}$	Quantifies computational cost per prediction, directly impacting deployment feasibility in production environments.

		Low resource requirements enable real-time threat forecasting at scale, whilst high costs may limit operational utility. Critical for assessing whether the model can be deployed in resource-constrained SOC environments or requires dedicated infrastructure.
Operational Readiness Index (ORI)	$0.3 \times \text{Precision} + 0.3 \times \text{Recall} + 0.2 \times (1 - \text{FPR}) + 0.2 \times \text{Resource_Efficiency}$	Composite metric measuring multiple operational concerns into a single readiness score. The weighted combination balances detection quality (precision, recall), operational burden (FPR), and practical constraints (resource efficiency). Provides stakeholders with a holistic assessment of whether the model is ready for production deployment and regulatory compliance (e.g., DORA requirements).